

Lecture 5 (9/23)

Expectation, Variance, and the Normal Distribution

Unit 1 Review

- Probability models: sample spaces, outcomes, events, and probability distributions
- Discrete random variables; joint, marginal, and conditional probabilities; Bayes' formula
- Independence and iid samples; simulation as an approximation to probability; translating between code and probability notation
- Continuous random variables: Uniform distribution, densities, CDFs

Goals for Unit 2:

- Expectation and variance, coefficient of variation
- Normal distribution; standardization and linear transformations
- The linear regression model assumptions
- Understand covariance and correlation

Sample averages and expectation

- The sample mean of iid draws $Y_1, \dots, Y_N$ of a variable with sample space S is

$$\bar{Y} = \text{np.mean}(y) = \frac{1}{N} \sum_{i=1}^N Y_i = \sum_{y \in S} y \frac{\#\{Y_i = y\}}{N}$$

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- For a continuous random variable,

$$E[Y] = \int_{-\infty}^{\infty} y f(y) dy$$

Expectation of a function

For any function g^1 of Y , we can similarly write

$$E[g(Y)] = \sum_{y \in S} g(y) P_Y(y) \quad \text{OR} \quad E[g(Y)] = \int g(y) f_Y(y) dy$$

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Taking $g(y) = 1_A(y)$ gives

$$E[1_A(Y)] = P(Y \in A).$$

A probability is just the expectation of an indicator.

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Example

Let Y take values in $\{1, 2, 3\}$ with $P(Y=1) = \frac{1}{2}$, $P(Y=2) = \frac{1}{3}$, $P(Y=3) = \frac{1}{6}$.
Compute $E[Y]$ and $E[Y^2]$.

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Example

Let $Y \sim \text{Uniform}(0, L)$. What is $E[Y]$? What is $E[Y^2]$?

Properties of expectation

- **Multiplication by a constant:** $E[aX] = aE[X]$
- **Factorization when independent:** If X and Y are independent, $E[XY] = E[X]E[Y]$.
- **Tower property:** $E[E[X | Y]] = E[X]$ where $E[X | Y]$ is the expectation of $X|Y$ (and hence depends on Y). This is just marginalization for expectations.

Example

Let

$$X \sim \text{Bernoulli}(1/3)$$

$$Y|X \sim \text{Uniform}(0, 1 + X)$$

Compute $E[Y]$.

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Variance and standard deviation

In addition to the expectation, we want to measure how spread out a distribution is. For this we define the variance

$$\text{var}(Y) = E[(Y - E[Y])^2] = E[Y^2] - E[Y]^2.$$

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Let $Y \sim \text{Bernoulli}(q)$. Compute $\text{var}(Y)$.

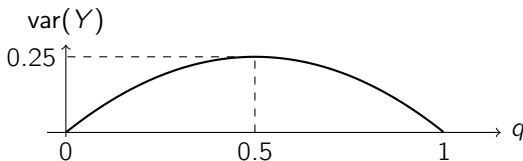
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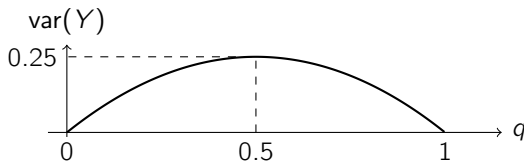
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Let $Y \sim \text{Bernoulli}(q)$. Compute $\text{var}(Y)$.



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Let $Y \sim \text{Uniform}(0, 1)$. Compute $\text{var}(Y)$.

Standard deviation

- The standard deviation is the square root of the variance,

$$\sigma = \sqrt{\text{var}(Y)} = \sqrt{E[(Y - E[Y])^2]}.$$

Unlike the variance, σ has the same units as Y , so it tells us the *typical size* of the residual $Y - E[Y]$.

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- The coefficient of variation, or CV , compares the standard deviation to the mean:

$$CV = \frac{\sigma}{\mu}$$

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Quantity	Typical CV
Adult height	≈ 0.05
Adult weight	≈ 0.2
Household income	≈ 1
Household wealth	3–5

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Note: The CV makes sense for positive random variables. For example, it is meaningless for something like temperature.

The Normal distribution

$Y \sim \text{Normal}(\mu, \sigma^2)$ has density

$$g(y) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(y-\mu)^2/2\sigma^2}, \quad E[Y] = \mu, \quad \text{var}(Y) = \sigma^2.$$

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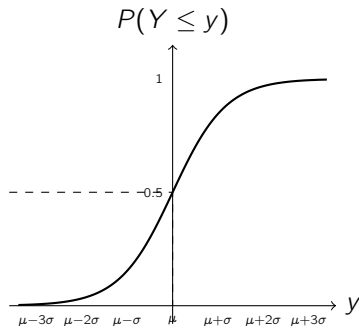
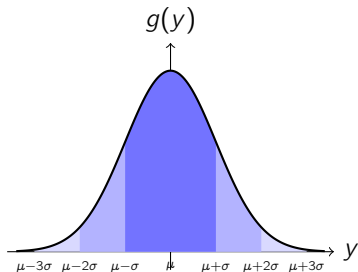
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Example

Suppose $Y \sim \text{Normal}(20, 25)$.

- Estimate $P(Y > 25)$ using the bell curve shape.
- Estimate $P(15 < Y < 30)$.

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Suppose $X_1 \sim \text{Normal}(3, 4)$ and $X_2 \sim \text{Normal}(5, 9)$ are independent, and $S = 2X_1 - X_2$.

- Find the distribution of S .
- Estimate $P(|S - 1| > 10)$.

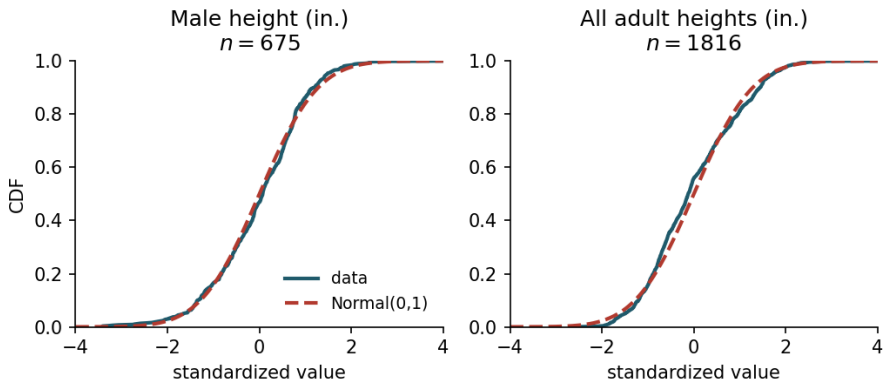
Plotting the CDF from samples

```
import numpy as np

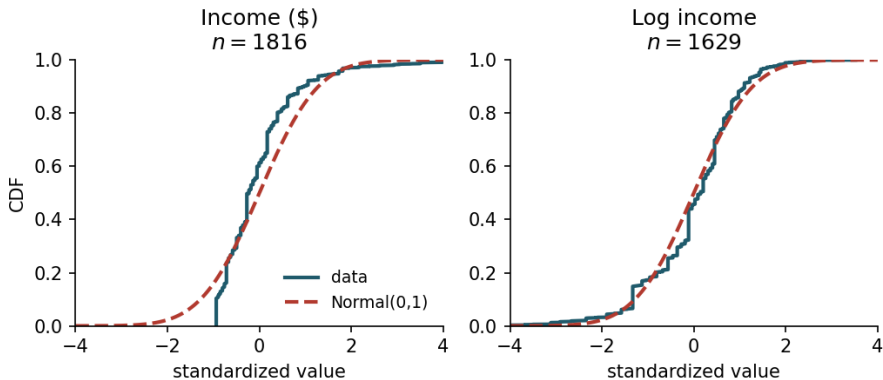
y = np.random.normal(0, 1, size=10000)

# CDF estimate at each threshold t
dt = 0.01
ts = [-4 + i * dt for i in range(800)]
cdf_hat = [len(y[y < t]) / len(y) for t in ts]

cdf_hat[i] estimates  $P(Y \leq ts[i])$ : the fraction of samples below that
threshold, exactly like the picture on the previous slide.
```



Standardized data vs. the standard Normal CDF (dashed), from a real survey:
avehtari/ROS-Examples, Earnings dataset.



Same dataset and method, for income and log income: raw income's CDF is far from the Normal curve (heavy right tail, plus a floor of respondents reporting \$0); log income is much closer.